

Automatic Authorship Attribution of Journalistic Texts Using Stylometric Features and Classical Machine Learning Methods

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Abstract

We compare word-based and explicit style representations for Reuters articles by William Kazer, Tim Farrand and Patricia Commins. All methods share 147 reference articles, three mystery texts and 150 official test articles. TF-IDF with nearest-neighbour attribution achieves 91.33% test accuracy, style features with nearest neighbours 69.33%, and style author centroids 82.67%. The experiment combines lexical exploration with a reproducible, matched evaluation. Case studies expose topic overlap and sensitivity to rare punctuation. The results support attribution for this fixed corpus and author selection, but do not establish topic-independent personal style.

1 Research question and scope

The project asks whether journalistic articles by three known authors can be distinguished using word patterns and explicit style measurements. We investigate two related questions: how well a word-based nearest-neighbour method performs beyond three illustrative mystery texts, and whether a small, interpretable set of style features also supports attribution.

The lexical exploration and the style comparison are complementary. Topic words can be useful predictors while also complicating claims about personal style. We therefore compare the representations on identical documents and report all specified methods. We use PCA and hierarchical clustering to inspect similarity. Classification uses the full feature spaces.

This is a compact course project, not an attempt to identify unknown writers across platforms or to build a general authorship detector. Federalist Papers provide a supplementary historical example. Reuters is the principal corpus, consistent with the project’s focus on journalism.

2 Data and held-out evaluation

Reuter_50_50 contains 50 authors with 50 training and 50 test articles per author [1]. We use William Kazer, Tim Farrand and Patricia Commins. These are the same authors used in the initial Reuters notebook.

Author	Reference	Mystery	Official test
William Kazer	49	1	50
Tim Farrand	49	1	50
Patricia Commins	49	1	50
Total	147	3	150

The alphabetically first filename in each author’s `C50train` folder is reserved as a mystery text. The remaining files form the reference set. The three mystery texts are not added back when evaluating `C50test`. Vocabulary, IDF, scaling, constant-feature removal, centroids and PCA use only reference articles. No model parameter or feature group was selected from the official test scores.

The corpus is appropriate for a journalistic attribution experiment, but the selected authors cover different topics and companies. Balanced counts do not control these differences. SHA-256 checks found no exact raw-byte or stripped-text duplicate between the reference set and either evaluation group. This does not rule out related stories or near duplicates.

3 Representations and decision rules

3.1 Word TF-IDF

A custom regular expression extracts words, hyphenated forms and currency expressions. The NLTK English Snowball stemmer processes word tokens [2]. The vectoriser uses English stop words, word n-grams of lengths one to three, at most 1,000 features, `min_df=2` and `max_df=0.9` [3]. The fitted vocabulary contains 832 unigrams, 151 bigrams and 17 trigrams. It is therefore not a vocabulary of 1,000 individual word stems.

For feature t in document d , the default smoothed weight before normalisation is

$$w_{d,t} = \text{tf}(t, d) \left[\log \left(\frac{1 + N}{1 + \text{df}(t)} \right) + 1 \right], \quad N = 147.$$

Each vector is then normalised to unit Euclidean length. The original English stop-word list and custom stemmer are not fully aligned, which produces a scikit-learn warning. We retain this pre-processing for the matched comparison and expose the warning rather than silently changing the representation.

3.2 Explicit style features

The spaCy model `en_core_web_sm` provides tokenisation and sentence boundaries [4]. Cleaning normalises line endings, decodes HTML entities and strips outer whitespace. Case and punctuation remain intact. Alphabetic tokens define words. We use 26 predefined measurements:

Group	Count	Measurements
Surface	5	Mean sentence length, sentence-length standard deviation, mean word length, type-token ratio and stop-word ratio
Function words	15	Frequencies of a, and, as, at, but, for, from, in, of, on, or, that, the, to, with
Punctuation	6	Frequencies of comma, period, colon, semicolon, question mark and exclamation mark

Frequencies are expressed per 1,000 alphabetic words. Sentence length counts alphabetic tokens in sentences that contain at least one such token. Punctuation counts exact spaCy punctuation tokens. Ratios reduce direct length dependence, but type-token ratio still depends on text length. POS n-grams are outside this fixed comparison.

The measurements have different units. Reference-set means and population standard deviations standardise each retained feature:

$$z_{d,j} = \frac{x_{d,j} - \mu_{j,R}}{\sigma_{j,R}}.$$

Exclamation-mark frequency is constant in the reference set and is removed. The remaining style vectors have 25 dimensions. Query documents never contribute to scaling. The implementation preserves the earlier calculation’s rounding conventions, documented in `REPRODUCIBILITY.md`.

3.3 Nearest neighbour and author centroid

For either representation, nearest-neighbour attribution finds the closest individual reference article:

$$\hat{y}(x) = y_{i^*}, \quad i^* = \arg \min_{i \in R} \|x - x_i\|_2.$$

The style-centroid method instead averages the reference style vectors of each author and selects the nearest author profile:

$$c_a = \frac{1}{|R_a|} \sum_{d \in R_a} z_d, \quad \hat{y}(z) = \arg \min_a \|z - c_a\|_2.$$

NumPy computes these distances and averages. The comparison tests the effect of the representation under the same nearest-neighbour rule, then the effect of averaging within the style representation. It does not isolate an independent causal contribution of each feature group.

4 Exploration with PCA and clustering

Reuters Ward clustering receives condensed Euclidean distances between the full TF-IDF reference vectors [5]. PCA is a separate display step, using full SVD fitted on the references [6]. Neither procedure changes the feature space used for the predictions.

Representation	PC1	PC2	Combined
TF-IDF, 1,000 dimensions	6.23%	5.35%	11.58%
Style, 25 dimensions	11.22%	9.71%	20.94%

Most variance lies outside the displayed plane. The larger style percentage does not imply better classification because the representations differ in dimension and variance structure. The matched plot in `results/matched_pca.png` shows the same reference documents and mystery stars in both spaces.

Author-average TF-IDF weights provide lexical context. The highest-weight terms include `china` for Kazer (0.1914), `million` for Farrand (0.1516), and `adm` for Commins (0.1231). These indicate topical or company-related differences. They are average TF-IDF weights, not PCA loadings or proof of a stable personal writing style.

The initial account reported slightly different values, including 6.20% for TF-IDF PC1 and 0.8650 for Farrand’s mystery-neighbour distance. The verified calculation gives 6.23% and 0.8615. The three mystery predictions agree. The cause of the small historical numerical differences has not been isolated. The individual lexical report preserves both sets of values.

5 Quantitative evaluation

Accuracy is the proportion of correct labels. Macro-F1 averages the three authors’ F1 scores with equal weight [7]. The official test set provides the primary evaluation.

Method	Mystery	Test correct	Accuracy	Macro-F1
TF-IDF, nearest neighbour	3/3	137/150	91.33%	0.9138
Style, nearest neighbour	1/3	104/150	69.33%	0.6864
Style, author centroid	2/3	124/150	82.67%	0.8283

Always predicting one author yields 33.33% accuracy on the balanced test. All three methods exceed this reference. TF-IDF performs best for this trio. Within the style representation, centroids outperform individual neighbours. This single comparison is consistent with a benefit from averaging, but does not establish a general advantage of centroid models.

The style-centroid confusion matrix is:

True author / predicted author	Kazer	Farrand	Commins
Kazer	40	8	2
Farrand	11	39	0
Commins	3	2	45

Recall is 80%, 78% and 90%, respectively. Most centroid confusion occurs between Kazer and Farrand. In contrast, ten of the thirteen TF-IDF errors assign Commins articles to Farrand. The other errors assign two Kazer texts to Farrand and one Farrand text to Kazer.

The three mystery documents are useful for demonstration, but their small sample changes the apparent ranking sharply. The style nearest-neighbour method gets only one mystery text correct while classifying 104 official test articles correctly. Three perfect TF-IDF mystery predictions likewise do not imply perfect test performance.

6 Qualitative analysis of individual decisions

We select three cases after evaluation, choosing the first sorted test path within each of three outcome categories. All happen to be Commins articles. These are illustrative cases, not a representative sample. We do not refit or tune models after inspection. Filenames and calculated diagnostics are provided in `results/qualitative_cases.json`.

Lexical error with correct style attribution. `C50test/PatriciaCommins/314014newsML.txt` reports Dayton Hudson store closures. Its nearest TF-IDF reference is Farrand’s `173770newsML.txt`, which discusses Burton’s retail profits. Shared features `store`, `retail`, `profit` and `divis` make the topic overlap explicit. Both style models correctly predict Commins. Her centroid is at distance 3.020, compared with 3.660 to Kazer and 3.729 to Farrand.

Correct lexical attribution with a centroid error. Commins’s `334989newsML.txt` discusses a possible write-down of Quaker’s Snapple business. Her reference article `255733newsML.txt` discusses a possible sale of the same business. `quaker` and `snappl` dominate their shared TF-IDF products. TF-IDF and style nearest neighbours are correct. The centroid predicts Kazer at distance 4.079, against 4.336 to Commins. The stop-word ratio and mean word length favour Kazer in the squared-distance comparison. Averaging does not capture every individual document’s style. The lexical success also relies on closely related subject matter.

Agreement on the wrong author. Commins’s `404426newsML.txt` describes CPC’s corn-refining business. All methods predict Farrand. The closest word-based reference is his `220852newsML.txt`, about Tate and Lyle and corn processing. Shared features include `corn`, `sweeten` and `syrup`. The centroid distances are 19.839 to Farrand and 20.030 to Commins. Semicolon frequency contributes 9.661 to Farrand’s squared-distance advantage, the largest absolute feature contribution. The article contains a product enumeration using that punctuation. A rare standardised feature can strongly affect the comparison.

For a centroid winner w and alternative a , each feature contributes $(z_j - c_{a,j})^2 - (z_j - c_{w,j})^2$ to the distance margin. Positive values favour the winner. These values describe the model calculation exactly, but are not causal explanations of writing style or global importance scores.

7 Implementation, checks and supplementary example

The repository has one main analysis entry point and small modules with explicit responsibilities:

File	Responsibility
<code>prepare_data.py</code>	Download the corpus and extract the selected articles
<code>run_analysis.py</code>	Assemble representations, evaluate methods and export evidence
<code>stylometry/corpus.py</code>	Load documents, assign roles and check exact duplicates
<code>stylometry/features.py</code>	Shared word tokenisation and fixed style measurements
<code>stylometry/models.py</code>	Reference scaling and NumPy nearest-neighbour distances
<code>stylometry/federalist.py</code>	Extract historical text bodies and separate label meta-data

A fresh Python 3.13.13 environment installed from the repository reproduced all earlier document identities, style measurements, predictions and model metrics. The package consistency check passed. Five semantic tests cover counting conventions, empty input, reference scaling, distance ties, cleaning and historical text parsing. All three notebooks ran in fresh kernels. Direct dependencies are pinned, the full Windows environment is recorded, and outputs contain relative corpus paths and hashes. Other operating systems have not been tested.

The course connection is concrete: NLTK and spaCy provide linguistic processing, pandas stores tables, NumPy computes distances, argparse supports file-path options, and CSV/JSON preserve outputs. Docstrings, small functions and a virtual environment support inspectability. The project does not require an artificial class hierarchy or every course topic.

The supplementary Federalist notebooks use Project Gutenberg eBook 18 [8]. A shared parser retains the first version of essay 70 and identifies essays 1–85 from headings. Titles, publication and author metadata, salutation and the PUBLIUS signature plus following endnotes are excluded from features. Essays 49–58, 62 and 63 form a disputed category, and 18–20 a joint category. The resulting groups contain 51 Hamilton, 14 Madison, 5 Jay, 12 disputed and 3 joint texts. The HCA uses word frequencies with IDF disabled and condensed Euclidean distances. The PCA uses TF-IDF. Their corrected plots replace earlier uncleaned exploration. They are descriptive examples, not a supervised Federalist evaluation.

8 Limitations and conclusion

Topic and author are not independently controlled. Even function words, sentence structure and punctuation can vary by subject matter and editorial practice. Related news across splits can increase lexical similarity. Exact-duplicate checks do not resolve these issues. Sentence boundaries have not been manually audited across all articles. Rare punctuation and text length can affect standardised style distances.

The fixed author selection and single official split limit generalisation. A useful extension would evaluate a separate author selection or a topic-controlled split. Feature-group ablation would require a clearly separated development process and an untouched evaluation set. We do not claim that such experiments were performed.

For the present Reuters trio, both representations contain useful attribution signals. TF-IDF achieves the strongest official-test result, while a small style-centroid model also performs substantially above the constant-label reference. The comparison and case studies show why this performance should be interpreted as corpus-specific evidence rather than proof of pure style.

Contributions and assistance

Kai S. Kurono’s project responsibility covers the Reuters and Federalist exploration notebooks, word representations, PCA, hierarchical clustering and Reuters mystery attribution. Jonaid Aydi’s responsibility covers the complementary style comparison, official-test evaluation, repository integration, reproducibility and bilingual documentation.

Codex assisted with integration, error diagnosis, verification and preparation of the reports and presentation. The two contributors remain responsible for the submitted work and its explanation.

References

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